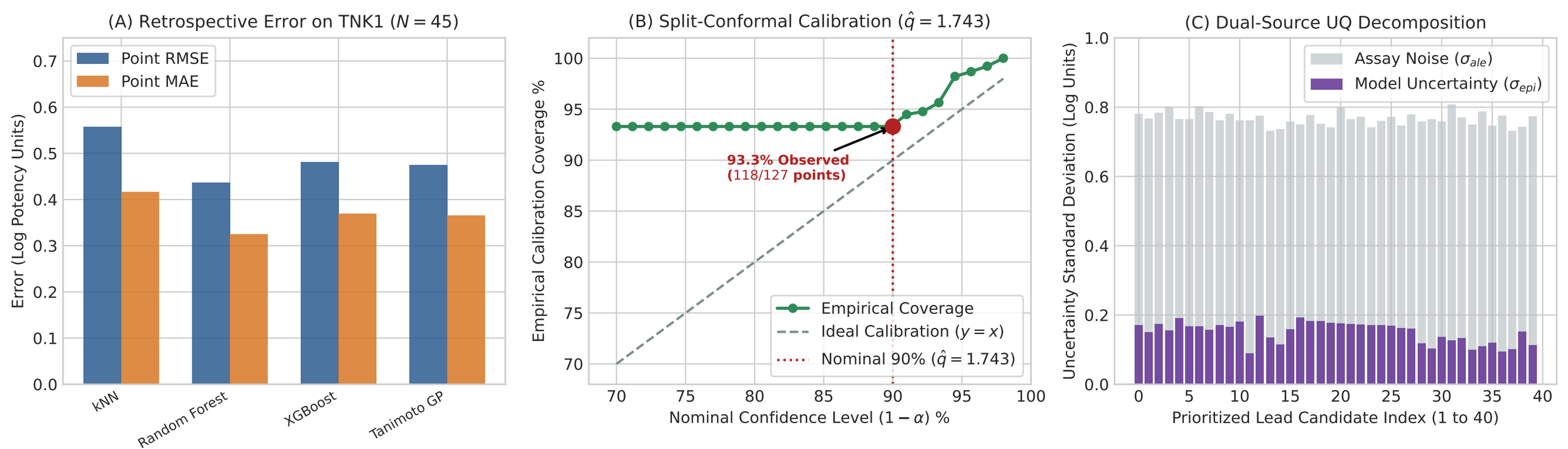
(A) Retrospective Error on TNK1 ($N = 45$)
Point RMSE
Point MAE
Error (Log Potency Units)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
kNN
Random Forest
XGBoost
Tanimoto GP
(B) Split-Conformal Calibration ($\hat{q} = 1.743$)
Empirical Calibration Coverage %
70
75
80
85
90
95
100
93.3% Observed
(118/127 points)
Empirical Coverage
Ideal Calibration ($y = x$)
Nominal 90% ($\hat{q} = 1.743$)
Nominal Confidence Level ($1 - \alpha$) %
(C) Dual-Source UQ Decomposition
Assay Noise (σ_{ale})
Model Uncertainty (σ_{epi})
Uncertainty Standard Deviation (Log Units)
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15
20
25
30
35
40
Prioritized Lead Candidate Index (1 to 40)